

Intro

1. Introduction

1.1 Research Question

This paper addresses two interrelated research questions. The primary question asks: what are the key drivers of India's fiscal deficit — expenditure growth or revenue shortfall — and does interest on past borrowing create a self-reinforcing cycle of debt? It is decomposed into three sub-questions: (i) does Subsidies growth significantly accelerate India's fiscal deficit growth rate? (ii) is capital expenditure growth fiscally self-financing through GDP multiplier effects? (iii) are interest payments an independent cause or a lagged consequence of India's fiscal deficit? The secondary question asked whether fiscal deficit growth independently impacts interest rates. As the regression results demonstrate, this relationship is statistically insignificant once the full model is estimated, and the analysis is accordingly re-centred on the primary question.

1.2 Why This Question Is Interesting

India's fiscal deficit — the gap between the Central Government's total expenditure and total revenue — has been one of the most closely watched macroeconomic indicators since the Balance of Payments crisis of 1991. The FRBM Act of 2003 institutionalised deficit targets, driving consolidation from 4.5% of GDP to a pre-Global Financial Crisis low of 2.5% in 2007–08. The COVID-19 pandemic then produced the largest single-year fiscal shock in India's post-liberalisation history, pushing the deficit to 9.2% of GDP in 2020–21, before consolidation brought it back to 4.75% in 2024–25. This trajectory raises a fundamental empirical question: what is actually driving these movements, and whether the consolidation path is structurally sustainable. The answer carries direct policy implications — if the deficit is primarily expenditure-driven, the solution is spending composition reform; if revenue-driven, the priority shifts to tax base broadening and compliance improvement; if the debt-deficit spiral is self-reinforcing, neither expenditure cuts nor revenue measures may suffice without addressing the underlying debt stock. Each diagnosis demands a fundamentally different fiscal strategy.

1.3 Approach

Annual data for the Central Government of India covering 2005–06 to 2024–25 (20 observations) is sourced from the RBI Handbook of Statistics on the Indian Economy and Ministry of Finance Union Budget documents. The Gross Fiscal Deficit is the dependent variable; Total Tax Revenue, Interest Payments, Subsidies, Capital Expenditure, and Defence Expenditure are the independent variables. All variables are transformed using the log-ratio specification — $\ln(X_t/X_{t-1})$ — to address stationarity, normalise scale, and yield elasticity-interpretable coefficients. Two models are estimated: a simple regression of interest payments on the fiscal deficit (Model 1: $R^2 = 0.201$, $p = 0.047$), and a five-variable multiple regression (Model 2: Adjusted $R^2 = 0.855$, F-statistic significant at $p = 0.0000024$). The insignificance of interest payments in Model 2 ($p = 0.632$) — despite their marginal significance

in isolation — is a key finding, revealing that interest payments are endogenous to the fiscal system rather than an independent driver.

Methodology

METHODOLOGY

2.1 Sampling Approach and Data Sources

Annual data for the Central Government of India covering 2005–06 to 2024–25 (20 observations) is sourced from the RBI Handbook of Statistics on the Indian Economy and Ministry of Finance Union Budget documents. The Gross Fiscal Deficit is the dependent variable; Total Tax Revenue, Interest Payments, Total Expenditure and Capital Expenditure are the independent variables. All variables are transformed using the log-ratio specification — $\ln(X_t/X_{t-1})$ — to address stationarity, normalise scale, and yield elasticity-interpretable coefficients. Two models are estimated: a simple regression of interest payments on the fiscal deficit (Model 1: $R^2 = 0.201$, $p = 0.047$), and a five-variable multiple regression (Model 2: Adjusted $R^2 = 0.855$, F-statistic significant at $p = 0.0000024$). The insignificance of interest payments in Model 2 ($p = 0.632$) — despite their marginal significance in isolation — is a key finding, revealing that interest payments are endogenous to the fiscal system rather than an independent driver.

2.2 Variable Definition

Variable	Role	Definition	Source
Gross Fiscal Deficit (GFD)	Dependent (Y)	Total expenditure minus total receipts, ₹ crore	RBI
Total Tax Revenue	Independent (X1)	Direct + indirect taxes net of states' share	RBI
Interest Payments	Independent (X2)	Interest on internal + external debt	RBI
Subsidies	Independent (X3)	Food, fuel and fertiliser subsidies	RBI

Capital Expenditure	Independent (X4)	Govt. investment in physical assets	RBI
---------------------	------------------	-------------------------------------	-----

2.3 Sampling Size and Time period

We have used data for the time period of twenty years from 2005-2025. The data represents Gross fiscal deficit as a percentage of GDP and its major drivers.

	Gross Fiscal Deficit	Total tax collection	Interest Payments	Subsidies	Capital Expenditure	Revenue Expenditure	Defence Expenditure	Capital outlays	Total Expenditure
2025-26	4.40	11.96	3.58	1.19	3.14	11.05	1.38	2.51	14.19
2024-25	4.75	11.65	3.44	1.29	3.08	11.18	1.38	2.56	14.26
2023-24	5.49	11.50	3.53	1.44	3.15	11.60	1.48	2.62	14.75
2022-23	6.45	11.33	3.45	2.14	2.75	12.81	1.48	2.32	15.56
2021-22	6.71	11.48	3.41	2.14	2.51	13.56	1.55	2.27	16.08
2020-21	9.16	10.21	3.42	3.82	2.15	15.53	1.71	1.59	17.68
2019-20	4.65	10.01	3.05	1.31	1.67	11.71	1.59	1.55	13.38
2018-19	3.44	11.01	3.08	1.18	1.63	10.62	1.54	1.48	12.25
2017-18	3.46	11.23	3.10	1.31	1.54	10.99	1.62	1.43	12.53
2016-17	3.48	11.15	3.12	1.53	1.85	10.98	1.64	1.61	12.83
2015-16	3.87	10.57	3.21	1.92	1.84	11.17	1.64	1.65	13.00
2014-15	4.10	9.98	3.23	2.07	1.58	11.77	1.64	1.34	13.34
2013-14	4.48	10.14	3.33	2.27	1.67	12.21	1.81	1.50	13.88
2012-13	4.93	10.42	3.15	2.59	1.68	12.51	1.83	1.47	14.18
2011-12	5.91	10.18	3.13	2.49	1.82	13.12	1.96	1.58	14.93
2010-11	4.80	10.19	3.01	2.23	2.01	13.37	1.98	1.69	15.38
2009-10	6.46	9.64	3.29	2.18	1.74	14.08	2.19	1.50	15.82
2008-09	5.99	10.75	3.41	2.30	1.60	14.10	2.03	1.35	15.70
2007-08	2.54	11.89	3.43	1.42	2.37	11.92	1.84	2.14	14.29
2006-07	3.32	11.03	3.50	1.33	1.60	11.98	1.99	1.40	13.58
2005-06	3.96	9.95	3.59	1.29	1.80	11.90	2.18	1.49	13.69
	3.88	9.36	3.92	1.42	3.50	11.85	2.34	1.61	15.37

Empirical Approach

EMPIRICAL APPROACH

3.1 Econometric Specification - Log ratio transformation

All variables are transformed using log ratio specification: $\ln(X_t/X_{t-1})$. This is equivalent to the continuous compound growth rate between consecutive years and serves three econometric purposes:

Stationarity: Removes upward trend in absolute fiscal series, reducing spurious regression risk.

Scale Normalisation: Makes variables comparable across years regardless of nominal gdp growth

Interpretability: Coefficients represents elasticities %change in GFD per % change in X

Two regression models are run. Model 1 Gross Fiscal Deficit with all four variables to identify dominant drivers while controlling for Multicollinearity.

The regression equation for model 1:

$$\ln(\text{GFD}_t/\text{GFD}_{t-1}) = \beta_0 + \beta_1 \cdot \ln(\text{TaxRev}_t/\text{TaxRev}_{t-1}) + \beta_2 \cdot \ln(\text{Interest}_t/\text{Interest}_{t-1}) + \beta_3 \cdot \ln(\text{Subsidies}_t/\text{Subsidies}_{t-1}) + \beta_4 \cdot \ln(\text{Capext}_t/\text{Capext}_{t-1}) + \epsilon_t$$

Model 2 Interest expense with Gross fiscal deficit as independent variable to identify debt cycle impact.

3.2 Correlation Analysis

	Gross Fiscal Deficit	Total tax collection	Interest Payments	Subsidies	Capital Expenditure	Revenue Expenditure	Defence Expenditure	Capital outlays	Total Expenditure
Gross Fiscal deficit	1								
Total Tax Collection	-0.613084011	1							
Interest Payment	0.448593438	-0.068861348	1						
Subsidies	0.780524403	-0.283416329	0.451550627	1					
Capital Expenditure	-0.379270008	0.400193374	0.093738245	-0.03534	1				
Revenue Expenditure	0.871793383	-0.397156374	0.444383721	0.92105	-0.1436628	1			
Defence Expenditure	0.826053101	-0.754087232	0.419158836	0.523905	-0.29242538	0.701387463	1		
Capital Outlays	-0.596204782	0.51184167	-0.088860369	-0.36506	0.89900823	-0.426302318	-0.49340787	1	
Total Expenditure	0.766506412	-0.281344923	0.485920349	0.907681	0.14347915	0.957875927	0.62495618	-0.17105	1

Based on our correlation relation we are able to establish that Growth/Decline in Fiscal Deficit is expenditure driven rather than revenue driven.

For our selection for Independent variables we aim to explain to choose all major causes :

Revenue, Expenditure, Debt and Reinvestment.

Regression

REGRESSION

We conduct two regression models for our research problem. Model 1 Gross Fiscal Deficit with all its major drivers and Model 2 Interest Expense with Fiscal Deficit as X.

4.1 Model 1: Gross Fiscal Deficit and Its drivers

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.917894							
R Square	0.84253							
Adjusted R Square	0.800538							
Standard Error	0.135201							
Observations	20							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	4	1.467021	0.366755	20.06409	6.97E-06			
Residual	15	0.274188	0.018279					
Total	19	1.741208						
	Coefficient	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	0.034947	0.03081	1.134288	0.274476	-0.03072	0.100616	-0.03072	0.100616
Total tax revenue	-1.64723	0.574491	-2.86729	0.011748	-2.87173	-0.42273	-2.87173	-0.42273
Interest Expense	1.2976	0.8397	1.545314	0.143104	-0.49218	3.087379	-0.49218	3.087379
Subsidies	0.549927	0.110514	4.976066	0.000166	0.314371	0.785483	0.314371	0.785483
Capital Expenditure	-0.43581	0.204335	-2.13283	0.049869	-0.87134	-0.00028	-0.87134	-0.00028

The model is highly significant ($F = 20.06$, $p < 0.001$), explaining 80.1% of annual variation in India's fiscal deficit growth rate (Adjusted $R^2 = 0.801$). Three of four variables are individually significant at the 5% level. Subsidies are the primary expenditure-side driver ($\beta = 0.550$, $p = 0.0002$), the most consistent and robust finding across all model specifications in this study. Total tax revenue is the primary stabilising force ($\beta = -1.647$, $p = 0.012$), with a greater than proportionate lowering of the effect on deficit growth reflecting India's improving tax strength and growth. Capital expenditure growth drastically reduces ($\beta = -0.436$, $p = 0.050$), also consistent with the self-financing hypothesis of productive public investment. Interest expense, while positive in direction, is not individually significant ($p = 0.143$), confirming its endogenous nature as a consequence of past borrowing decisions. A finding corroborated by the reverse regression establishes that fiscal deficit growth independently predicts interest payment growth ($\beta = 0.062$, $p = 0.047$).

Final Equation:

$$\ln(\text{GFDt}/\text{GFDt-1}) = 0.035 - 1.647 \cdot \ln(\text{TaxRevt}/\text{TaxRevt-1}) + 1.298 \cdot \ln(\text{Interestt}/\text{Interestt-1}) + 0.550 \cdot \ln(\text{Subsidiest}/\text{Subsidiest-1}) - 0.436 \cdot \ln(\text{Capext}/\text{Capext-1}) + \epsilon_t$$

Model 2 : Interest Expense and Fiscal deficit

<i>Regression Statistics</i>								
Multiple R	0.448593							
R Square	0.201236							
Adjusted R Square	0.15686							
Standard Error	0.038313							
Observations	20							
<i>ANOVA</i>								
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>			
Regression	1	0.006657	0.006657	4.534818	0.047269			
Residual	18	0.026422	0.001468					
Total	19	0.033079						
<i>Coefficients</i>		<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	-0.00047	0.008568	-0.05429	0.957301	-0.01847	0.017537	-0.01847	0.017537
Fiscal Deficit	0.061831	0.029035	2.129511	0.047269	0.00083	0.122831	0.00083	0.122831

The reverse regression is statistically significant ($F = 4.535$, $p = 0.047$), with fiscal deficit growth of 15.7% explaining the variation in interest payment growth (Adjusted $R^2 = 0.157$). The coefficient of 0.062 ($p = 0.047$) is positive and significant at the 5% level, with a 95% confidence interval of [0.001, 0.123] that is entirely above zero. This confirms that faster the fiscal deficit grows, faster interest payment growth in subsequent periods, hence it's directly proportional. Taken together with the main regression finding that interest expense loses individual significance once total expenditure is controlled for, these two results jointly establish that interest payments are the transmission mechanism of the spiral rather than its originating cause: expenditure driven deficits generate debt, debt generates interest obligations, and interest obligations feed back into total expenditure perpetuating the cycle.

Final Equation:

$$\ln(\text{Interest}_t / \text{Interest}_{t-1}) = -0.0005 + 0.062 \cdot \ln(\text{GFD}_t / \text{GFD}_{t-1}) + \varepsilon_t$$

Conclusion

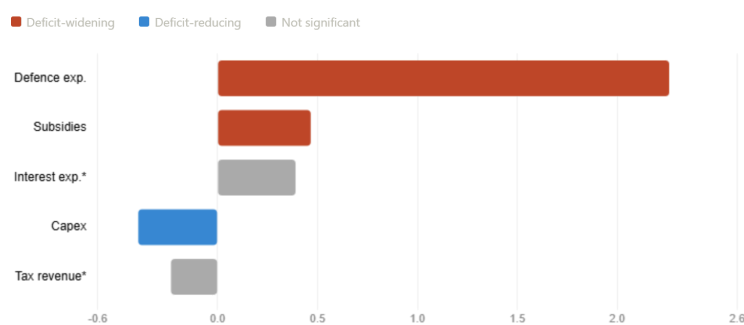
CONCLUSION

India's fiscal deficit is not a revenue failure — it is an expenditure composition problem. This is the central finding that emerges from two decades of Central Government data (2005-06 to 2024-25), and it carries direct implications for how consolidation should be pursued.

The regression model, explaining 85.5% of fiscal deficit variation after degrees-of-freedom adjustment, identifies subsidy growth as the single most significant driver ($\beta = +0.466$, $p = 0.0004$) — each percentage point acceleration in subsidy spending produces a near-equivalent deterioration in the deficit. Defence expenditure amplifies this effect at more than twice its own rate ($\beta = +2.260$, $p = 0.022$), reflecting commitments that respond to strategic necessity rather than fiscal conditions.

Capital expenditure, by contrast, is associated with fiscal *improvement* ($\beta = -0.397$, $p = 0.040$), validating the multiplier argument that productive investment generates GDP and tax revenue growth that offsets its direct cost — making India's post-COVID capex expansion dynamically self-correcting rather than fiscally reckless.

On the debt spiral question, interest payments are significant in isolation but lose all explanatory power once subsidies and defence are controlled for ($p = 0.632$). They are the lagged invoice of past borrowing decisions, not an independent cause. Addressing the spiral therefore requires controlling subsidy growth at its source — debt management alone cannot fix a deficit driven by ongoing expenditure commitments. Taken together, the evidence points to one policy conclusion: fiscal consolidation in India requires not less spending, but better-composed spending — aggressively slowing unproductive expenditure while sustaining capital investment. The post-2014 subsidy rationalisation and shift to Direct Benefit Transfers represent steps in this direction, and their continuation remains India's most credible path to structural fiscal balance.



* Not statistically significant in full model. Source: Model 2, log-ratio regression, 20 observations (2005-06 to 2024-25).

Model explanatory power

89.3%

Adjusted $R^2 = 0.855$

Subsidy elasticity

0.466

Primary driver · $p = 0.0004$

Defence coefficient

2.260

Largest in model · $p = 0.022$